

Lecture 08

PH 101

OUTLINE

- Generalized co-ordinates
- Degrees of freedom and constraints

Let's start with ultra-basics!

- One of the fundamental concept of mechanics is that of a **particle**.

What do you mean by a 'particle'?

***By this we mean a body whose dimensions may be neglected in describing its motion.**
The possibility of so doing depends, of course, on the conditions of the problem concerned.
For example, the planets may be regarded as particles in considering their motion about the Sun, but not in considering their rotation about their axes.

**Mechanics, Landau and Lifshitz*

AS YOU KNOW!

The position of a particle in space is defined by its radius vector $\vec{r}$, whose components are its Cartesian co-ordinates x, y, z .

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \text{ or } \vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

The derivative $\vec{v} = \frac{d\vec{r}}{dt}$ of $\vec{r}$ with respect to the time t is called the velocity of the particle, and the second derivative $\frac{d^2\vec{r}}{dt^2}$ its acceleration.

□ We are going to denote differentiation with respect to time by placing a dot above a letter:

$$v = \dot{r}$$

- To define the position of a **single** particle in space, it is necessary to specify just one radius vectors, i.e. 3 co-ordinates: (x, y, z)
- To define the position of **two particles moving freely, without any constraints, in space**, it is necessary to specify two radius vectors, i.e. 6 co-ordinates: (x_1, y_1, z_1) and (x_2, y_2, z_2)

To define the position of a system of N particles moving freely in space, without any constraints, it is necessary to specify N radius vectors, i.e. $3N$ coordinates:

$$(x_1, y_1, z_1; x_2, y_2, z_2; \dots \dots; x_N, y_N, z_N)$$

- The **number of independent quantities** which must be specified in order to define uniquely the position of any system is called the number of **degrees of freedom**; here, this number is $3N$.
- These quantities need not be the Cartesian co-ordinates of the particles, and the conditions of the problem may render some other choice of coordinates more convenient.

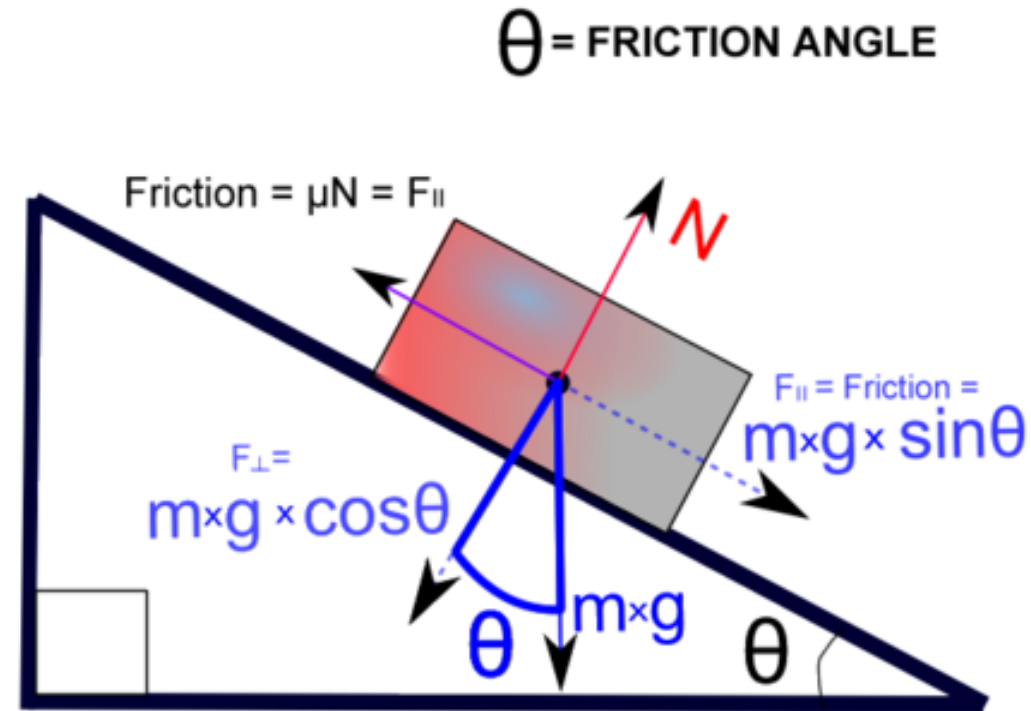
Example:

A single particle moving on a plane can be represented by the coordinates (x, y)
if we use **Cartesian coordinate system**

Or by the coordinates (r, θ) if **plane-polar coordinate system** is used.

What we mean by a constraint?

A **constraint** on a system is a parameter that the system must obey. For example, a box sliding down a slope must remain on the slope.

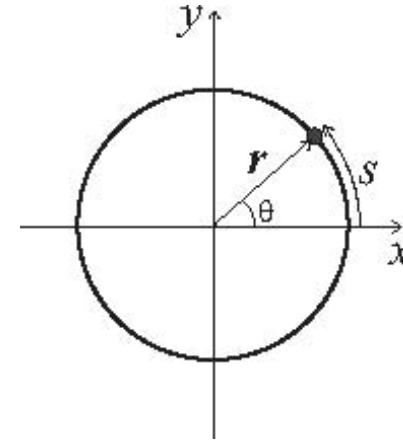
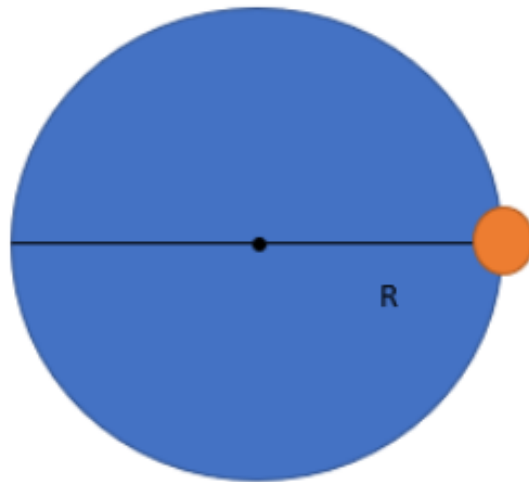


Source: Wikipedia

Fig. 1: In this system the box slides down a slope, the constraint is that the box must remain on the slope (it cannot go through it or start flying).

- A **constraint** puts limitations to the motion of a system and **the number of independent coordinates needed to describe the motion is reduced.**

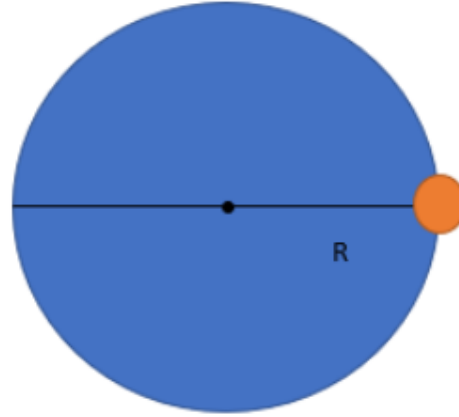
Example 1: A particle allowed to move on the circumference of a circle of radius R
(Using the Plane-polar coordinate system)



- Only one coordinate $q_1 = \theta$ is sufficient to describe the motion of the particle. Because the radius of the particle remains the same.
- The constraint equation is $|\vec{r}| = R$

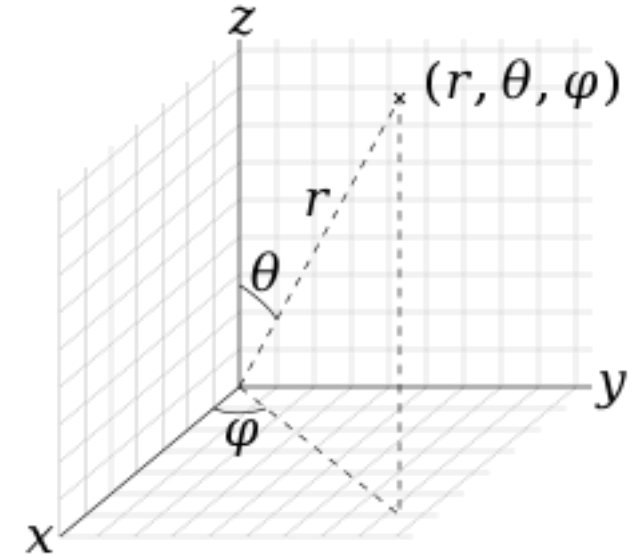
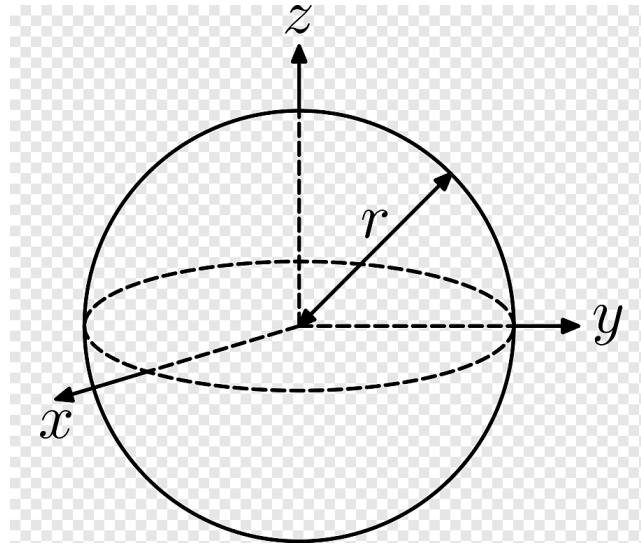
Example 2: A particle allowed to move on the circumference of a circle of radius R

(Using the Cartesian coordinate system)



- The constraint equation here is: $x^2 + y^2 = R^2 = \text{constant}$
- Clearly because of the constraint equation, **we have only one independent variable: x or y**
- So, only one independent variable is sufficient to describe the motion.
- Also it is evident that the degrees of freedom=1

Example 2: A particle constrained to move on the surface of a sphere.



- Clearly, the constraint equation is: $r = R$
- The most suitable coordinate system to use is the Spherical polar coordinate system: r, θ, ϕ
- Because of the constraint, r cannot vary, it is fixed. Only θ and ϕ can vary.
- So there are only two independent coordinates: $q_1 = \theta$ and $q_2 = \phi$
- Also it is evident that the degrees of freedom=2

Holonomic constraints

- Consider a system of N particles. Say, **constraints** are present in the system.
- If the constraints are expressed in the form of equations of the type: $f(\vec{r}_1, \vec{r}_2, \dots, t) = 0$
- Such constraints are termed as **holonomic constraints**.

Say, there are m number of such equations to describe the constraints in the N particle system.

We can use these equations to eliminate m of the $3N$ coordinates and we will then need only n independent coordinates to describe the motion, where $n = 3N - m$

The system is said to have n or $3N - m$ degrees of freedom.

The elimination of the dependent coordinates can be expressed by introducing $n = 3N - m$ independent variables $q_1, q_2, \dots, q_n$. **These are referred to as the generalized coordinates.**

More about them shortly!

Superfluous Coordinates : The idea of degrees of freedom makes it clear that when we are using, say rectangular cartesian coordinates, we have several *redundant* or *superfluous* coordinates, if there are holonomic constraints. This redundancy and non-independence of the coordinates makes the problem complicated and this difficulty is resolved by using the generalized coordinates. For example, let us consider a body be thrown vertically upward with an initial velocity v_0 . The body will move in a straight line. In cartesian coordinates, the motion will be represented as

$$x = 0, \quad y = v_0 t - \frac{1}{2} g t^2, \quad z = 0$$

where X and Z axes are horizontal and Y -axis is in vertical direction. At different values of the time t , only y coordinate varies and x and z coordinates remain the same. Therefore x and z coordinates are *superfluous coordinates*. In conclusion, we need only one coordinate $q = y$ to describe the vertical motion.

Very quickly about: **Nonholonomic constraints**

If the constraints **CANNOT** be expressed in the form of equations of the type: $f(\vec{r}_1, \vec{r}_2, \dots, t) = 0$, they are termed as **nonholonomic** constraints.

Example:

A particle placed on the surface of a sphere in the presence of a gravitational field

In this case the particle is able to fall off the sphere under the influence of gravity.
The constraint equation can be put in the form:

$$|\vec{r}| \geq R \quad \text{or} \quad r - R \geq 0$$

We will not bother much about nonholonomic constraints in this course. 😊

Now, as we know, in the absence of any constraints:

To define the position of a system of N particles moving freely in space it is necessary to specify N radius vectors, i.e. $3N$ coordinates:

$$(x_1, y_1, z_1; x_2, y_2, z_2; \dots \dots; x_N, y_N, z_N)$$

What about the case, if there exist constraints?

- If there exists equations of **constraint** that relate some of these coordinates to others, then not all the $3N$ coordinates are independent.
- In fact, if there are m equations of constraints, then $(3N - m)$ coordinates are independent.
- And the system is said to possess $n = (3N - m)$ degrees of freedom.

Some worked out examples

Example 1

Determine the number of degrees of freedom for six particles moving freely in a plane.

Answer:

Each free particle needs two coordinates to specify its position in a plane. So, six free particles will need 12 coordinates. There are no constraints. Hence the number of degrees of freedom of the system is 12.

Example 2

Determine the number of degrees of freedom for two particles connected by a rigid rod moving freely in a plane.

Answer:

The two particles in a plane will need 4 coordinates: (x_1, y_1) and (x_2, y_2) . The constraint in the system is that the distance l between the two particles is fixed.

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 = l^2 = \text{constant}$$

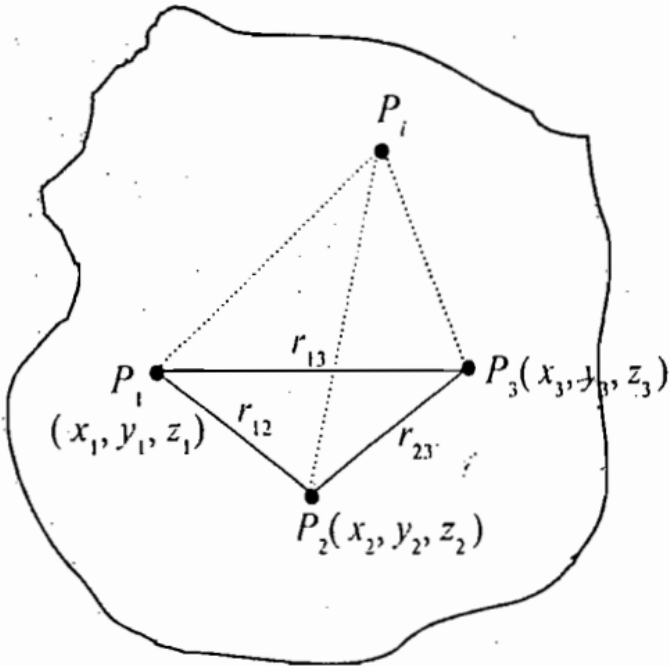
So the degrees of freedom = $2 \times 2 - 1 = 3$

Example 3

Determine the number of degrees of freedom for a rigid body moving freely in space (3D).

Answer:

A rigid body is a system of particles in which the distance between any two particles remain fixed throughout the motion. If the body is rigid, it is sufficient to describe the position of at least three non-collinear particles. This makes it possible to reconstruct the position of all the other particles, provided that their time-invariant position relative to the three selected particles is known.



Let us consider three non-collinear particles P_1, P_2 and P_3 in the rigid body. As each particle has 3 degrees of freedom and there are 3 constraints of form:

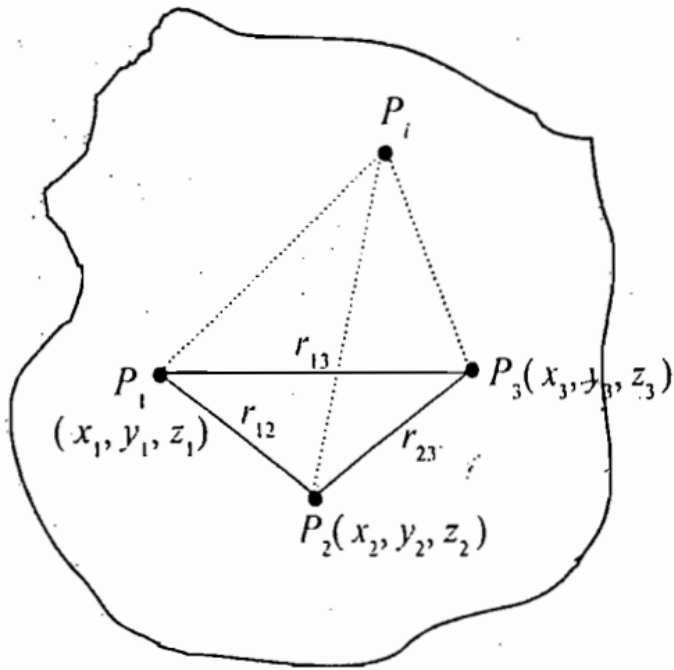
$$(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 = r_{12}^2$$

$$(x_3 - x_2)^2 + (y_3 - y_2)^2 + (z_3 - z_2)^2 = r_{32}^2$$

$$(x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2 = r_{13}^2$$

Clearly, the degrees of freedom these particles P_1, P_2 and P_3 are:

$$3N - m = 3 \times 3 - 3 = 6$$



NOW, consideration of any other particle, say P_i , needs three coordinates (x_i, y_i, z_i) . Moreover, it is obvious that there are three equations of constraints, because the distances of P_i from P_1, P_2 and P_3 are fixed. **Hence** any other particle will **NOT add** any new degrees of freedom to the six degrees of freedom of the three-particle system of the body. Thus **a rigid body moving freely in three-dimensional space has 6 degrees of freedom.**

Generalized coordinates

If the configuration of a system S is determined by the values of a set of independent variables $q_1, \dots, q_n$, then $\{q_1, \dots, q_n\}$ is said to be a set of generalized coordinates for S .

- (i) When we say the generalised coordinates must be **independent variables**, we mean that there must be *no functional relation connecting them*. If there were, one of the coordinates could be removed and the remaining $n - 1$ coordinates would still determine the configuration of the system. The set of generalised coordinates must not be reducible in this way.
- (ii) When we say the generalised coordinates $q_1, \dots, q_n$ **determine the configuration** of the system S , we mean that, when the values of the coordinates $q_1, \dots, q_n$ are given, the position of every particle of S is determined. In other words, the position vectors $\{\mathbf{r}_i\}$ of the particles must be known functions of the independent variables $q_1, \dots, q_n$, that is,

$$\mathbf{r}_i = \mathbf{r}_i(q_1, \dots, q_n) \qquad (i = 1, \dots, N).$$

Understanding generalized coordinates via EXAMPLES!

Example 1

Let S be the system shown in Figure 2 which consists of two particles P_1 and P_2 connected by a light rigid rod of length a . The particle P_1 is constrained to move along a fixed horizontal rail and the system moves in the vertical plane through the rail. **Select generalized coordinates for this system and obtain expressions for the position vectors r_1, r_2 in terms of these coordinates.**

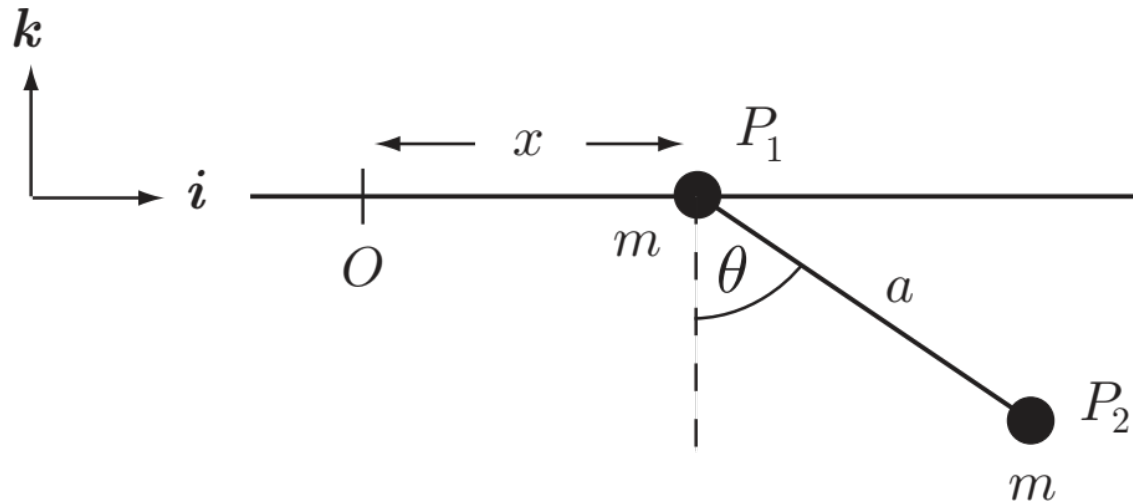
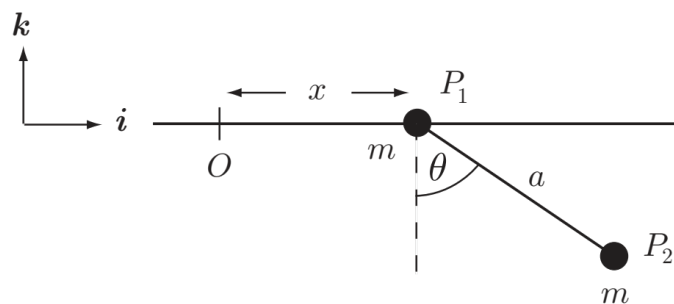


Fig.2

Solution:



Consider the variables x , θ shown. These are certainly independent variables (they are not connected by any functional relation) and, when they are given, the configuration of $\mathcal{S}$ is determined. Thus $\{x, \theta\}$ is a set of **generalised coordinates** for the system $\mathcal{S}$.

In terms of the coordinates x and θ , the positions of the particles P_1 and P_2 are given by

$$\mathbf{r}_1 = x \mathbf{i},$$

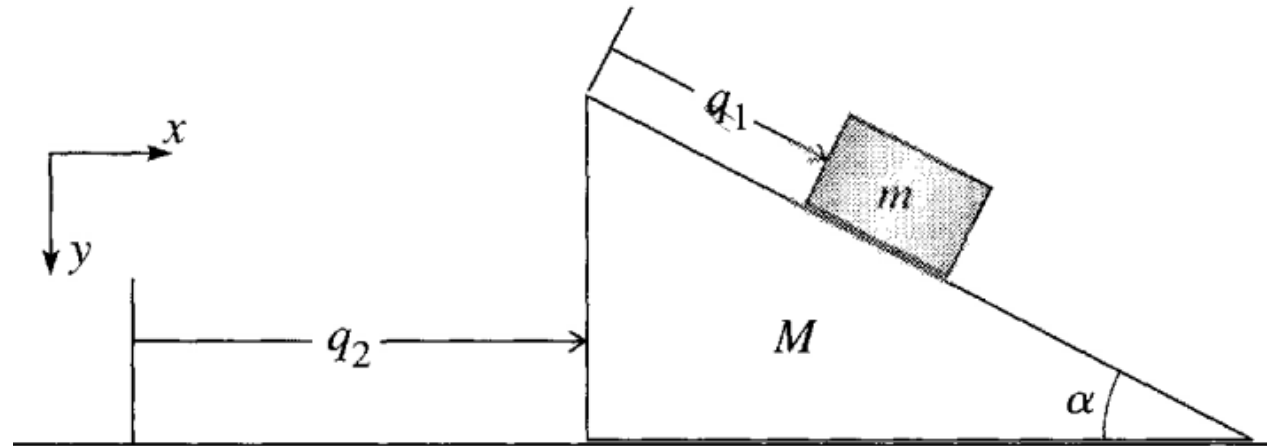
$$\mathbf{r}_2 = (x + a \sin \theta) \mathbf{i} - (a \cos \theta) \mathbf{k},$$

Example 2

Consider the block and wedge shown in Figure 3. The block (mass m) is free to slide on the wedge, and the wedge (mass M) can slide on the horizontal table, both with negligible friction. The block is released from the top of the wedge, with both initially at rest. The wedge has angle α and the length of its sloping face is l . If you are asked to find how long does the block take to reach the bottom*, what would be your choice for the generalized coordinates?

Answer:

The system has two degrees of freedom, and a good choice of the two generalized coordinates is, as shown, the distance q_1 of the block from the top of the wedge and the distance q_2 of the wedge from any convenient fixed point on the table.



*We will solve this problem later using Lagrangian formalism

